Proposal for Reorienting the VES Pipeline Around Ancient Greek Uncial Data

A technical concept note for pilot-scale evaluation

Working draft for technical review. Prepared in support of evaluation of a bounded computational pilot and possible future use of research computing resources.

Abstract

This document outlines a proposal to redirect the current the current VES (a machine-learning pipeline for ink detection and character- oriented reconstruction on virtually unwrapped ancient Greek material) toward a more historically and paleographically appropriate training regime for ancient Greek uncial material. The existing codebase contains a functioning segmentation and classification workflow, but parts of its present design appear to reflect assumptions inherited from synthetic or modern-handwriting-oriented data rather than the morphology and textual context of ancient Greek papyri. The central claim of this proposal is that closer alignment between training data, preprocessing choices, and the actual target domain may improve the usefulness of the system for Vesuvius-related work. This document summarizes the current system, identifies several technical concerns, explains why the `ALPUB_v2` uncial dataset is especially important, and proposes a bounded pilot plan for further evaluation.

1. Purpose of This Document

This document presents a technical rationale for reorienting the current VES pipeline around ancient Greek uncial data. Its immediate purpose is not to claim that the existing approach has already succeeded, but to show that there is a plausible and testable path forward that is better aligned with the visual and textual realities of the source material.

The VES repository already contains a substantial machine-learning workflow: a synthetic-data pipeline, a segmentation and classification model, training code, and visualization and inference tools. The project is therefore not at the stage of initial speculation or empty scaffolding. The present question is whether its assumptions are sufficiently well matched to the domain it is intended to address.

My working hypothesis is that the system can be improved by replacing or subordinating earlier assumptions derived from synthetic or modern Greek handwriting data and by grounding the pipeline more directly in ancient Greek uncial forms and their likely textual environment. This document explains the reasons for that hypothesis and proposes a concrete direction for a pilot evaluation.

More broadly, the aim is to clarify which elements of the current approach remain promising, which elements appear to reflect legacy assumptions that no longer fit the target domain well, and which changes are both technically and historically defensible. The proposal is therefore diagnostic as well as constructive: it attempts to identify where the present pipeline may be misaligned, while also pointing toward a path that can be tested through staged experimentation.

2. Current State of the Pipeline

The existing VES codebase implements a custom machine-learning workflow aimed at ink detection and character-oriented reconstruction tasks on virtually unwrapped Greek material. In broad terms, the repository currently includes:

- a synthetic data-generation pipeline for producing papyrus-like input imagery
- a segmentation and classification model built around a hybrid attention-based architecture
- a training loop for grayscale single-character inputs and associated masks
- visualization and inference scripts for inspecting outputs

This represents genuine technical progress. The project already contains a functional training pipeline, a nontrivial model architecture, and a dataset interface capable of supporting further experimentation. That is important because it means the next stage of work can focus on domain alignment and evaluation rather than basic implementation.

At the same time, the repository appears to preserve some assumptions from an earlier stage of development. Those assumptions may have been reasonable while the project was still oriented toward synthetic or more generic Greek-script workflows, but they now need to be re-evaluated in light of the actual target domain: ancient Greek uncial letterforms preserved in papyrological or related manuscript imagery.

The practical significance of this point is straightforward. If the model architecture is already serviceable, then improvement may depend less on inventing a new system from scratch than on supplying the present system with better-matched data, better-matched preprocessing, and more defensible training assumptions. In other words, the current codebase should be understood not as a failed endpoint, but as a usable base for a more domain-appropriate line of inquiry.

3. Central Technical Problem

The central problem is domain mismatch.

The current workflow appears to have inherited assumptions from synthetic or modern-handwriting-oriented data that do not map cleanly onto ancient Greek uncials as they appear in papyrological material. Even when the labels are nominally “Greek,” the underlying morphology, stroke behavior, tonal structure, spacing conventions, and textual context can differ substantially.

This matters because machine-learning systems are shaped not only by model architecture but by the structure of the data used to train them. If the training regime emphasizes modern pen-written forms, modern punctuation expectations, or preprocessing steps that suppress visual information needed for ancient material, then the system may become optimized for the wrong problem.

There is also a textual dimension to the mismatch. The likely content of the Herculaneum villa corpus is not arbitrary Greek prose. It is more plausibly associated with philosophical and related material than with authors chosen merely as convenient benchmarks. That does not mean other authors are useless as tests, but it does mean that both visual priors and textual priors should be selected with greater care if the goal is to support historically plausible reconstruction.

For that reason, this proposal argues that the next phase of work should be organized around a narrower and more defensible question: whether a pipeline trained and evaluated on data that more closely reflects ancient Greek uncial forms can outperform, or at least more meaningfully inform, the present approach.

The remainder of this document identifies the main reasons for concern in the current setup, explains why the `ALPUB_v2` uncial corpus is especially important, and proposes a concrete line of investigation that can be tested through a bounded computational pilot.

4. Main Concerns

4.1 Domain Mismatch

The most important concern is domain mismatch. Ancient Greek uncials written with a reed pen or comparable broad-nib instrument do not have the same stroke structure, contrast profile, or range of letter variation as modern Greek minuscule handwriting written with ballpoint pens. Even where labels overlap at the level of alphabetic identity, the morphology can differ substantially.

A modern Greek handwritten text will often include modern line spacing, paragraph structure, monotonic accents, and punctuation conventions shaped by post-classical editorial practice. By contrast, much of the readable Herculaneum material is closer to *scripta continua*, often lacking spacing and accents, though punctuation and related marks may appear in some cases. Even where accentuation is present in ancient texts, it belongs to a polytonic system rather than the modern monotonic standard. If the target is to improve interpretation of ancient Greek material, then training assumptions should be anchored as closely as possible to ancient letterforms and their textual realities rather than to “Greek script” in a broad and undifferentiated sense.

4.2 Preprocessing Assumptions

Manual preprocessing experiments suggest that no single global recipe works reliably across all samples. In particular, hard thresholding and aggressive cleaning can destroy useful grayscale structure. Some samples remain legible only when tonal information is preserved rather than collapsed into a binary representation.

This point is consistent with papyrological practice more broadly. Different renderings of the same image can make different textual features more or less legible, which suggests that preprocessing should be treated as conditional and data-sensitive rather than as a one-size-fits-all transformation. In practical terms, this means that preprocessing is not merely a preliminary housekeeping step. It is part of the research problem itself. A representation that improves edge contrast may help one class of traces, while a grayscale-preserving representation may better preserve faint tonal differences that would be lost under hard thresholding.

4.3 Class Distribution

The extracted uncial corpus is not evenly distributed across letters. Some classes are abundant, while others are sparse. That imbalance likely reflects a combination of genuine language frequency, preservation bias, annotation bias, and scribal variation. For that reason, equalizing the dataset itself by force may be less defensible than preserving the real corpus and compensating during training through weighting or sampling strategies.

4.4 Geometry and Input Assumptions

The earlier active pipeline operated with a fixed `128x128` image size and a `32x32` mask size. The extracted `ALPUB_v2` uncial images are natively `70x70` RGB images. This does not imply that training must occur at exactly `70x70`, but it does suggest that the older geometry may reflect a legacy of the earlier workflow more than a principled fit to the uncial source data.

Accordingly, image and mask geometry should be treated as an empirical design choice to be tested rather than inherited as fixed doctrine from a previous stage of development.

4.5 GCDB Uppercase Assumptions

The training code appears to treat the GCDB-based source, that is, the modern Greek character dataset used in an earlier stage of the project, as an uppercase-only character set, regardless of how the underlying dataset may originally have been organized. In the generator, folder labels are normalized with `.upper()`, and the active class map is written as a single uppercase-style label set rather than as distinct uppercase and lowercase classes.

That simplification may have arisen once it was recognized that the target text is uncial rather than minuscule. Even so, modern Greek capitals written with modern pens are not an adequate substitute for ancient uncial forms written with ancient tools. The issue is therefore not simply case normalization. It is the broader assumption that modern uppercase material can stand in for ancient uncial morphology without substantial loss of relevance.

5. Why the ALPUB_v2 Dataset Matters

The extracted `ALPUB_v2` corpus is important because it provides a far more relevant source of ancient Greek character imagery than modern handwriting datasets. It contains 205,797 total samples across 24 letter classes, uses native `70x70` RGB images, preserves historically relevant forms such as lunate sigma, and exhibits a naturally uneven but informative class distribution.

For present purposes, the importance of the dataset is not merely quantitative. Its value lies in the fact that it supplies a more plausible visual basis for training assumptions. Even if it introduces complications in balancing, preprocessing, and label handling, it is a much stronger starting point for domain-aligned experimentation than datasets derived from modern Greek handwriting.

6. Revised Direction of Work

The next stage of work should focus on a staged experimental program built around historically better-matched uncial data, preprocessing that preserves useful grayscale structure, and training methods that compensate for imbalance without flattening the corpus itself. The objective is not immediate finalization, but disciplined evaluation of whether better domain alignment yields better behavior.

In practical terms, this means re-centering the data pipeline on ancient uncial material, preserving real class distributions while compensating during training, grouping train and validation splits by document rather than by isolated sample, and treating preprocessing as an experimental variable rather than a fixed precondition. It may also require renewed attention to label design where paleographically meaningful variants are currently being collapsed too aggressively.

A further change in the revised pipeline is the move toward manifest-based training on the ALPUB_v2 corpus rather than continued reliance on the older egacy path-pair format. This makes the dataset representation more explicit, preserves document-level metadata, and provides a cleaner basis for filtering, grouping, and staged experimentation. In practical terms, the training workflow is now better aligned with a corpus of ancient Greek uncial images treated as a coherent dataset rather than as a loosely assembled collection of isolated path pairs.

The revised training procedure also uses document-level splitting for training and validation partitions. This is an important methodological change, because it reduces the risk that highly similar samples from the same source document appear in both partitions, which would make validation results appear stronger than they really are. Grouping by document does not solve every evaluation problem, but it does produce a more defensible experimental setup than unconstrained random sample splitting.

In the revised training setup, the class distribution presented during training is also mildly rebalanced at the sampler level. Manifest-based records are sampled with inverse-square-root weighting by document frequency, which increases exposure to rarer letters while avoiding the stronger distortions that would result from full inverse-frequency balancing. The intent is not to erase the underlying corpus distribution, but to reduceoverrepresentation of the most common classes and make the training signal somewhat less skewed.

The reconstruction task itself has also been reframed for manifest-based data. Instead of relying only on pre-existing noisy/clean path pairs, the current pipeline degrades clean uncial images on the fly to generate controlled inputs while preserving the original clean image as the reconstruction target. This creates a more explicit corruption-and-recovery setup and allows the degradation process to be tuned without changing the underlying clean corpus.

Finally, labels are normalized to a canonical Greek-letter inventory at the metadata level before training. Variant names are mapped into a common label space so that the model is not asked to learn distinctions introduced merely by inconsistent naming in the source metadata.

At the same time, this normalization must be applied with caution. Sigma forms are a good example: visual inspection suggests that most samples in the LunateSigma class are genuinely lunate, but some appear to resemble positional sigma forms closer to what later became initial or medial lowercase sigma, while others resemble *sigma periestigmenon* or related editorial signs. This distinction matters. Positional sigma forms remain phonetic sigmata, but they also encode structural information about position within a word or syllable, whereas editorial signs such as *sigma periestigmenon* are non-phonetic. A training pipeline that collapses all such forms too casually risks erasing information that may be paleographically and textually significant.

7. Why a Pilot on Research Computing Resources Is Justified

The present state of the project suggests that a small, carefully bounded pilot on research computing infrastructure is justified.

The reason is not that the project has already proven its larger claims. It has not. Rather, the reason is that the project has moved past the stage of pure speculation and into a stage where a limited amount of serious compute would allow several important questions to be answered decisively. A local proof-of-concept now exists, the training pipeline runs, and the principal remaining obstacle is the cost of iteration at meaningful scale.

At present, local CPU-based testing is sufficient for smoke tests, debugging, and very small staged runs. It is not sufficient for efficient model comparison or timely evaluation on larger subsets of the corpus. Full runs on commodity local hardware are likely to take many hours or days, which sharply limits the number of experiments that can be performed and makes it difficult to assess whether changes in data curation, preprocessing, geometry, or model configuration materially improve results.’

A pilot on research computing resources would therefore serve a narrow and practical purpose: not to operationalize the project at final scale, but to determine whether the proposed reorientation toward ancient Greek uncial data is strong enough to merit further investment.

The initial pilot could be structured conservatively. A first run would use a single modern GPU on one node, train for one epoch on a limited subset, and produce checkpointed outputs, timing data, and loss curves. If that run succeeds, a second run could scale to a larger subset or a full one-epoch benchmark. These results would answer several questions that cannot be addressed efficiently on current local hardware:

- whether the present pipeline ports cleanly to a modern GPU environment
- whether the revised data assumptions produce stable training at larger scale
- whether runtime is low enough to support real experimental iteration
- whether output quality is promising enough to justify a larger allocation or extended collaboration

Just as importantly, a negative result from such a pilot would still be useful. If the model fails to scale well, if the revised data assumptions do not improve behavior, or if output quality remains weak even under better compute conditions, then the project can be narrowed, revised, or discontinued on the basis of evidence rather than conjecture. In that sense, the pilot is a filtering mechanism as much as an enabling mechanism.

For these reasons, the most reasonable next step is a modest pilot allocation suitable for benchmarking, reproducibility, and initial evaluation. The deliverables from such a pilot would be concrete:

- one or more reproducible training runs
- runtime and resource-usage measurements
- saved checkpoints and sample outputs
- a short report assessing whether the approach appears viable at larger scale

That is the level at which this project should presently be judged. The proposal is not that the underlying problem has already been solved, but that the project has matured enough to warrant a disciplined pilot on proper research computing infrastructure.

8. Conclusion

This proposal does not claim that the present approach has already solved the relevant technical problems. It argues instead that the project has reached a stage at which those problems can be stated more clearly, tested more directly, and evaluated more efficiently than before. The existing codebase, the availability of more relevant uncial data, and the initial success of local proof-of-concept runs together justify a bounded pilot rather than either premature dismissal or premature large-scale commitment.

The immediate task is therefore to test the revised direction under conditions that permit meaningful iteration and measurement. If the pilot demonstrates stable training, reasonable throughput, and outputs that are directionally encouraging, then a stronger case can be made for broader collaboration and more sustained use of research computing resources. If it does not, the project will at least have reached a clearer and more defensible conclusion than would be possible through continued small-scale local experimentation alone.

9. Appendix A. Local Validation Summary

Initial local validation was performed to determine whether the revised VES pipeline could be executed reliably in the present environment and whether local CPU hardware was sufficient for staged experimentation. These tests were conducted after correcting earlier issues involving shell-script robustness, interpreter selection, and smoke-test execution settings.

A smoke test completed successfully in CPU mode using /usr/bin/python3 and PyTorch 2.8.0+rocm6.4. This run used 64 samples, 1 epoch, batch size 4, disabled visualization, disabled TensorBoard, and a reduced smoke-test 0configuration designed to minimize instability. It completed successfully and saved both an intermediate checkpoint and a final model:

Epoch 1/1 | train_loss=7.1367 val_loss=7.0424

- Saved model for level 0 -> ./checkpoints//0-1.pt

- Saved model to ./new.pth

A larger staged local CPU run was then attempted to evaluate whether the revised pipeline remained practical beyond minimal smoke-test scale. That run used 2048 samples, 1 epoch, batch size 8, and the same corrected local execution path. It did not fail for environmental reasons, but it exceeded the configured 20 minute hard timeout:training on cpu

training levels: [0]

Training level: 0

training with 2048 items

On step 1

Smoke test still running after 600s

Smoke test hit hard timeout after 1200s

Taken together, these results support a limited but important conclusion. The revised pipeline is now locally executable and stable enough for debugging and minimal validation, but local CPU hardware becomes impractically slow, even at moderate subset scale. This strengthens the case for a bounded GPU-based pilot on research computing infrastructure.

10. References

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